Modeling Housing Price Dynamics in Melbourne: Using Regression Techniques

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Introduction

Objective: The primary objective of this project is to thoroughly explore, analyze and model factors that influence housing prices in Melbourne. The study integrates a wide range of variables, including numerical features, categorical factors, additive and interaction terms to develop a comprehensive understanding of the real estate market dynamics. Emphasis is placed on examining how key property characteristics such as number of rooms, number of bathrooms, bedrooms, distance and region etc. can impact the valuation of residential properties. By applying statistical modelling and variable selection techniques, the project aims to identify significant predictors, reveal complex interactions, and provide actionable insights into the drivers of housing prices in Melbourne's diverse property market.

Research Questions:

1. Which model would be best to predict house prices?
2. How do the number of rooms and distance from the city impact housing prices?
3. Do changing in property type shows different type of pricing pattern?
4. How will the price change with respect to the number of bathrooms?
5. Is there a notable interaction between room count and property region?

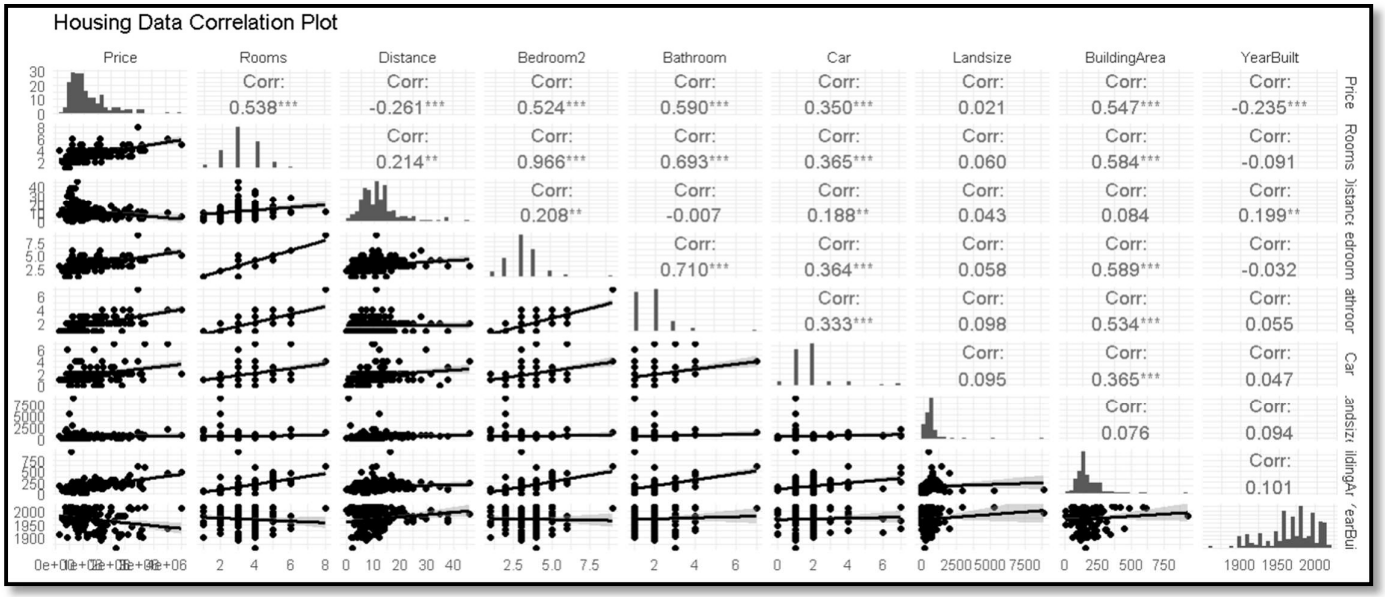
Hypothesis:

- Houses command a price premium over units and townhouses, even after controlling rooms and location.
- Being in the Southern Metropolitan region increases prices, especially for houses.
- The distance on price effect is nonlinear – properties close to mean distance have exponentially high price range.
- The marginal price increase per additional room decreases beyond 4 bedrooms.

Data Description:

- **Numerical Variables:** Property price, number of rooms, number of bathrooms, number of bedrooms, parking spaces, building area (in square meters), land size, year built, and distance from Melbourne's Central Business District (CBD).
- **Categorical Variables:** Type of property (house, townhouse, unit/apartment) and property region (e.g., Northern Metropolitan, Southern Metropolitan.
- **Interaction Terms:** Various interaction effects were created between major variables to capture more complex relationships influencing price dynamics (e.g., interaction between rooms and region, building area and bathrooms).

Exploratory Data Analysis:



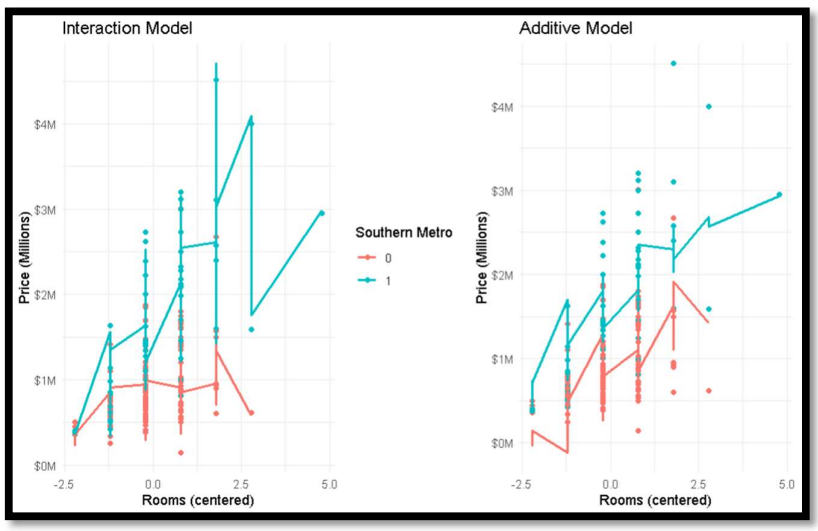
- Variables such as the number of rooms (Corr = 0.538), number of bathrooms (Corr = 0.590), building area (Corr = 0.547), and car parking spaces (Corr = 0.350) show strong positive correlations with housing prices, suggesting larger and better-equipped houses are priced higher.
- Distance from the city center is negatively correlated with price (Corr = -0.261), indicating that properties closer to the Central Business District (CBD) tend to be more expensive.
- Some variables like rooms and bedroom counts (Corr = 0.966), and bathrooms and bedrooms (Corr = 0.710) are highly correlated among themselves, highlighting a good possibility to use those variables as interaction terms.

```
Call:
lm(formula = Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X22 + X111 + X222 + X1111 + X2222 + X1X2 + X1X3
X1X4 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 + X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 + X3X4 +
X3X5 + X3X6 + X3X7 + X3X8 + X3X9 + X3X10 + X4X5 + X4X6 + X4X7 + X4X8 + X4X9 + X4X10 + X5X6 + X5X7 + X5X8 + X5X9 +
X5X10 + X6X7 + X6X8 + X6X9 + X6X10 + X7X8 + X7X9 + X7X10 + X8X9 + X8X10 + X9X10 + X126 + X145 + X235, data = data)

Residuals:
Min      1Q  Median      3Q     Max
-810040 -124006  -16566  121194  917699

Coefficients: (3 not defined because of singularities)
(Intercept) -3.126e+07  6.104e+06  -5.121 1.00e-06 ***
X1 -5.809e+06 1.179e+06 -4.928 2.34e-06 ***
X2 -5.728e+05 1.928e+05 -2.970 0.003510 **
...
X235 -2.518e+04  4.756e+04  -0.529 0.597415

Residual standard error: 297700 on 138 degrees of freedom
Multiple R-squared:  0.8826, Adjusted R-squared:  0.8307
F-statistic: 17.01 on 61 and 138 DF,  p-value: < 2.2e-16
```

- The multiple linear regression model achieved a strong fit with an R-squared of 0.8826
- The model is highly statistically significant with an F-statistic of 17.01 on 61 and 138 degrees of freedom and a p-value < 2.2e-16, confirming that the predictors jointly have a meaningful impact on house prices.
- The residuals show a relatively wide range suggesting the presence of some outliers or variability not fully captured by the model.

```
> anova(model1, model2)
Analysis of Variance Table

Model 1: Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 + X1X2
+
    X1X3 + X1X4 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 +
    X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 +
    X3X4 + X3X5 + X3X6 + X3X7 + X3X8 + X3X9 + X3X10 + X4X5 +
    X4X6 + X4X7 + X4X8 + X4X9 + X4X10 + X5X6 + X5X7 + X5X8 +
    X5X9 + X5X10 + X6X7 + X6X8 + X6X9 + X6X10 + X7X8 + X7X9 +
    X7X10 + X8X9 + X8X10 + X9X10 + X126 + X145 + X235
Model 2: Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X11 + X22 + X111 + X222 +
X1111 +
    X2222
  Res.Df        RSS  Df   Sum of Sq    F    Pr(>F)
1    144 1.4318e+13
2    187 3.9329e+13 -43 -2.5011e+13 5.85 4.964e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
```

- An ANOVA test was performed to compare the interaction model (Model 1) with the additive model (Model 2).
- The F-statistics of 5.85 and extremely small p-value (4.964e-16) indicate that the interaction model provides a significantly better fit than the simpler additive model.
- Including interaction terms between predictors significantly improves the model's ability to explain variations in housing prices, highlighting the importance of considering variable interactions.



- In the interaction model, the effect of the number of rooms on house price varies significantly between Southern Metro and other regions, showing different slopes for each group.
- The additive model assumes a consistent effect of rooms on price across locations, reflected by parallel lines with a constant slope but different intercepts.
- The interaction model is more flexible and better captures complex relationships, while the additive model is simpler and assumes uniform price increases per additional room.
- When modeling house prices, incorporating interaction terms can better reflect real-world differences across regions, leading to more accurate predictions.

Regression Analysis

Model Selection:

Backward Elimination:

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 299900 on 171 degrees of freedom
Multiple R-squared:  0.8524,    Adjusted R-squared:  0.8283
F-statistic: 35.28 on 28 and 171 DF,  p-value: < 2.2e-16
```

```
Step:  AIC=5166.8
Y ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X22 + X2222 +
    X1X2 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 + X2X3 +
    X2X4 + X2X6 + X3X5 + X5X6 + X6X7 + X6X8 + X6X9 + X6X10 +
    X7X10
```

Forward Selection:

```
Residual standard error: 324500 on 182 degrees of freedom
Multiple R-squared:  0.8161,    Adjusted R-squared:  0.7989
F-statistic:  47.5 on 17 and 182 DF,  p-value: < 2.2e-16
```

```
Step:  AIC=5093.2
Y ~ X5X10 + X7 + X2X6 + X3X7 + X1111 + X7X10 + X22 + X1X7 + X2X3 +
    X222 + X2X4 + X5X6 + X8 + X4X10 + X2X8 + X2X9 + X5X8
```

Stepwise Regression:

```
Residual standard error: 284400 on 160 degrees of freedom
Multiple R-squared:  0.8758,    Adjusted R-squared:  0.8456
F-statistic: 28.94 on 39 and 160 DF,  p-value: < 2.2e-16
```

```
Step:  AIC=5058.61
Y ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X22 + X111 +
    X1111 + X2222 + X1X2 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 +
    X1X10 + X2X3 + X2X4 + X2X6 + X2X8 + X3X5 + X3X9 + X4X5 +
    X4X7 + X4X10 + X5X6 + X5X9 + X5X10 + X6X7 + X6X8 + X6X9 +
    X6X10 + X7X10 + X8X9 + X126
```

Model Comparison:

- **Backward Elimination**: Achieved an R^2 of 0.85 but had the highest AIC (5166.8), suggesting a relatively good fit but less optimal model complexity.
- **Forward Selection**: Produced a lower AIC (5093.2) but also a lower R^2 (0.81), indicating the model is simpler but explains less variance in housing prices.
- **Stepwise Regression**: Outperformed both methods with the lowest AIC (5058.61) and the highest R^2 (0.87), making it the best balance between model fit and complexity.
- **Model Comparison Insight**: Lower AIC values indicate better models, and higher R^2 values indicate better explanatory power — Stepwise regression excels in both.
- **Conclusion:** Stepwise regression was selected as the final model due to its superior performance in both minimizing AIC and maximizing R^2, ensuring a more accurate and efficient prediction model.

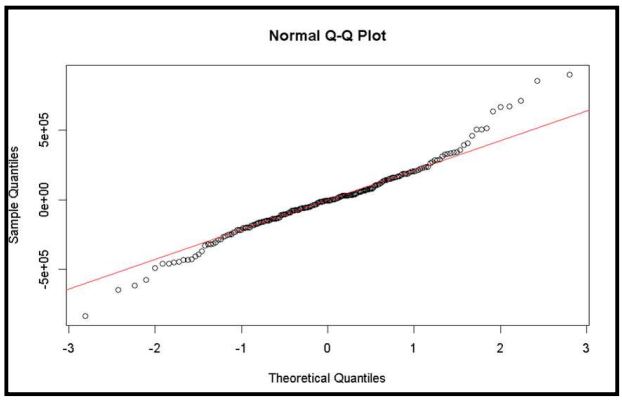
Why Stepwise Model: Stepwise regression was selected because it achieved the **lowest AIC (5058.61)** and the highest **R^2 value (0.87)** among all compared methods. This indicates that the stepwise model offers the best balance between model simplicity (AIC) and explanatory power (measured by R^2). It efficiently identified a subset of variables and interactions that significantly impact housing prices without overfitting.

Interesting Findings:

- **Rooms and Bathrooms**: Strong positive coefficients suggest that adding rooms and bathrooms significantly increases property value, especially when considered in larger or well-located homes.
- **Distance from City**: A negative coefficient was observed, confirming that houses located farther from Melbourne's central business district tend to have lower prices.
- **Interaction Terms**: Several interactions between variables like "Rooms × Region" indicated that the value added by extra rooms varies depending on the property's region — particularly more pronounced in areas like the Southern Metropolitan region.

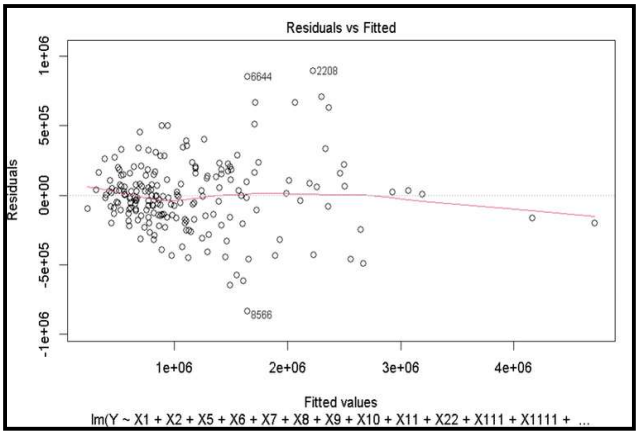
Residual Analysis:

Normality Assumption Check:



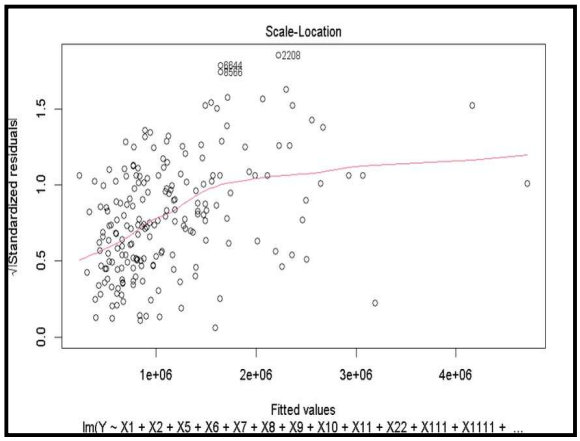
Good Normality in the Center: The residuals closely follow the reference red line within the central region (approximately between -1 and +1 on the standardized scale), indicating that for most of the data, the assumption of normality is well met.

Linearity Assumption Check:



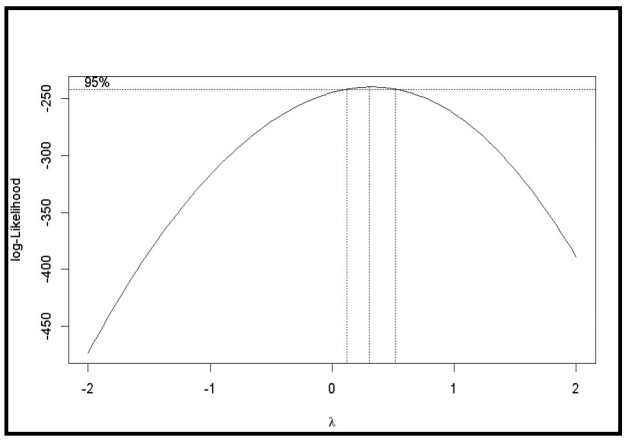
There is a mild violation of the linearity assumption, as some deviation from a perfectly linear relationship is observed. Additionally, heteroscedasticity is present, with residuals spreading out more as fitted values increase, indicating non-constant variance. The presence of influential outliers may also be affecting the model's overall performance and stability.

Homoscedasticity Assumption Check:



The residual plot indicates the presence of heteroscedasticity, as the variability of residuals increases with higher fitted values. There is also evidence of a heavy lower tail, with negative residuals deviating below the expected line. Additionally, potential outliers are observed, which could influence the model's accuracy.

Box-Cox Plot Check:



Based on the Box-Cox plot, the optimal λ is approximately 0. So, we have chosen logarithmic transformation of the dependent variable Y to stabilize variance and better satisfy model assumptions.

Model Fit Analysis (Stepwise Log Model):

```
Residual standard error: 0.2687 on 160 degrees of freedom
Multiple R-squared:  0.8224,    Adjusted R-squared:  0.7791
F-statistic:    19 on 39 and 160 DF,  p-value: < 2.2e-16
```

- **Strong Overall Fit**: The model achieves a high Multiple R-squared value of 0.8224, indicating that approximately 82% of the variance in the log of housing prices is explained by the selected predictors and their interactions.
- **Good Adjusted R-squared**: The Adjusted R-squared is 0.7791, suggesting that the model remains strong even after adjusting for the number of predictors, and that overfitting is reasonably controlled.
- **High Model Significance**: The overall F-statistic (F = 19, p-value < 2.2e-16) confirms that the model is highly significant, meaning at least some predictors reliably explain variation in the target variable.
- **Residual Behavior**: The residual standard error is 0.2687, which is reasonably low considering the log transformation applied, although mild issues like heteroscedasticity and influential outliers were noted in residual diagnostics and should be considered when interpreting predictions.

Discussions & Limitations

Room Count and City Distance Impact on Prices:

```
> summary(stepwise_model_log)$coefficients[c("X1", "X2", "X11", "X22", "X111", "X1111",
"X2222", "X1X2"), ]
              Estimate   Std. Error    t value     Pr(>|t|)
X1     -4.629126e+00 1.029349e+00 -4.4971375 1.317045e-05
X2     -1.538372e-01 1.630494e-01 -0.9435005 3.468475e-01
X11     1.122526e+00 3.260590e-01  3.4427081 7.351004e-04
X22     1.583681e-03 5.069777e-04  3.1237685 2.120098e-03
X111   -6.679791e-03 1.625187e-02 -0.4110168 6.816101e-01
X1111   1.842377e-02 5.379054e-03  3.4250946 7.809302e-04
X2222  -3.931247e-07 5.139549e-07 -0.7649011 4.454571e-01
X1X2   -4.273476e-02 5.150888e-02 -0.8296583 4.079681e-01
```

- **Room Count Effect**: The number of rooms shows a non-linear relationship with housing prices. Initially, adding more rooms appears to lower the price slightly, possibly due to inefficiencies or smaller home layouts. However, beyond a certain point, additional rooms significantly contribute to increasing the property value.
- **Distance Impact**: The distance from the city center has a subtle influence on prices. After a particular threshold, greater distance is associated with a slight increase in price, though overall, distance alone is not observed to be a strong or consistent predictor of housing prices in this dataset.
- **No Interaction Between Room Count and Distance**: There is no significant interaction between the number of rooms and distance from the city. This suggests that the effect of having more rooms on price is relatively independent of how far the property is located from the city center.

Impact of Property Type on Pricing Patterns:

```
> summary(stepwise_model_log)$coefficients[c("X2X3", "X2X4", "X3X5", "X3X9"), ]
          Estimate   Std. Error    t value   Pr(>|t|)
X2X3  0.0436574622 0.0170504433  2.5604884 0.01137763
X2X4  0.0417022758 0.0172120041  2.4228600 0.01651502
X3X5 -0.4023592446 0.1569515268 -2.5635892 0.01128056
X3X9  0.0001498087 0.0001802642  0.8310508 0.40718338
```

- **Property Type Matters**: The type of property plays a significant role in determining housing prices. Units and townhouses exhibit different pricing behaviors depending on the location, emphasizing the importance of considering property type when modeling and predicting prices.
- **Southern Metro Pricing Trends**: In the Southern Metropolitan region, units tend to be priced noticeably lower compared to both houses and townhouses. This suggests that buyers in this area may place a higher value on larger or detached properties relative to smaller unit-style dwellings.
- **Land Size Impact on Units**: Land size appears to have a diminished effect on the pricing of units when compared to houses and townhouses. This indicates that for unit properties, other factors—such as location and internal features—play a more dominant role than land area in determining market value.

Bathroom Count and Its Effect on Prices:

```
> summary(stepwise_model_log)$coefficients[c("X7", "X6X7", "X7X10"), ]
          Estimate   Std. Error   t value     Pr(>|t|)
X7    -7.867762106 1.6398560048 -4.797837 3.649902e-06
X6X7   2.385851018 0.5021355237  4.751409 4.466213e-06
X7X10  0.002060728 0.0005803872  3.550608 5.049832e-04
```

- **Bathroom Count Increases Value in Larger Homes**: Adding more bathrooms tends to significantly boost property value, particularly in larger, well-equipped homes that already have ample living space and bedrooms. In these cases, additional bathrooms are seen as enhancing comfort and luxury.
- **Potential Negative Impact in Smaller Homes**: In smaller properties, adding extra bathrooms can sometimes decrease the home's value. This may happen because additional bathrooms reduce the usable living space or are perceived as an inefficient use of limited square footage.
- **Bathroom Upgrades Are Most Effective with Size**: Improvements or additions to bathrooms are most beneficial when combined with spacious floor plans and multiple bedrooms. Buyers appear to value bathroom enhancements more when they complement a home's overall size and functionality.

Room Count and Property Region Interaction:

```
> summary(stepwise_model_log)$coefficients[c("X1X5"), ]
     Estimate    Std. Error       t value      Pr(>|t|)
-4.107591e+00  8.355547e-01 -4.916005e+00  2.170652e-06
```

- **Diminished Impact of Extra Rooms in Southern Metro**: In the Southern Metropolitan region, adding additional rooms to a property does not lead to as large a price increase compared to other regions. Buyers seem less responsive to increases in room count alone.
- **Smaller Price Gains for Added Rooms**: Properties located in the Southern Metro area experience a smaller boost in price when extra rooms are added. This suggests that simply increasing the room count is not a strong value driver in this region.
- **Buyers in Southern Metro Prioritize More Than Space**: Homebuyers in Southern Metro appear to prioritize overall quality, design, or other amenities over merely having more rooms. As a result, just expanding the number of rooms may not meet buyer expectations.

Prediction Analysis: The predicted price for the first house is approximately **$1,005,247**, based on the model's mean prediction. Although the 95% prediction interval ranges from **$578,410 to $1,747,068**, it provides a broad and realistic range that captures the natural variability in house prices, ensuring that most future observations are likely to fall within this interval.

Improvements in Analysis:

- Since the prediction interval is wide, we could explore using more refined models (e.g., random forests, gradient boosting) or applying stratification (analyzing separate models for different property types) to reduce variability.
- Since residual spread increases with fitted values, we might consider weighted least squares (WLS) or transformation techniques to correct for heteroscedasticity and stabilize variance.
- Investigate and possibly mitigate the impact of influential outliers through diagnostics like Cook's distance or leverage plots, ensuring they don't disproportionately affect model estimates.

While the current model provides strong explanatory power, several enhancements could further strengthen the analysis. Incorporating out-of-sample validation techniques, such as cross-validation, would give a more realistic picture of model performance. Addressing heteroscedasticity using weighted least squares or variance-stabilizing transformations could improve reliability.

Reliability of Data:

- **Reliability**: The data is generally reliable, with key property features consistently recorded, though minor inaccuracies (e.g., in land size measurements) could exist.
- **Validity**: The dataset captures essential factors influencing housing prices, ensuring good content validity.
- **External Validity**: Since the data is specific to Melbourne regions, predictions may not generalize well to other cities or markets.
- **Limitations**: Factors like interior quality, recent renovations, or neighborhood reputation are not captured, slightly limiting predictive accuracy.

Conclusion

Project Summary: In this project, we explored and modeled the factors influencing housing prices in Melbourne by combining various numerical, categorical, and interaction variables. We focused particularly on features like room count, number of bathrooms, property type, distance from the city, and property region. Using a stepwise selection approach, we built a predictive model that captures complex relationships, including significant interactions between key features. The model showed a strong overall fit, although some heteroscedasticity and mild non-linearity were observed. Through interpretation of coefficients and prediction analysis, valuable insights about Melbourne's housing market dynamics were derived.

Final Points:

- **Room Count**: More rooms generally increase price, but the effect varies by region and property size.
- **Bathroom Impact**: Additional bathrooms significantly boost value in larger, better-equipped homes.
- **Property Type**: Units and townhouses show different pricing patterns, especially in Southern Metro.
- **Interaction Effects**: Relationships between variables like room count and location are crucial, not all effects are additive.
- **Model Performance**: The final model achieved high R-squared and reasonable predictive ability, despite some residual variability.

Appendix

Data Set: Source: https://www.kaggle.com/datasets/anthonypino/melbourne-housing-market

Sample Size: 200 randomly selected observations from 8887 observations.

Code:

```r
data1 <- read.csv("C:/Users/satya/Downloads/Melbourne_housing_FULL_cleaned.csv",
                  header=TRUE)
set.seed(123)
data <- data1[sample(nrow(data1), 200), ]
attach(data)
names(data)
View(data)


library(ggplot2)
library(plotly)

Y <- Price
data$Rooms_c <- data$Rooms - mean(data$Rooms, na.rm = TRUE)
data$Distance_c <- data$Distance - mean(data$Distance, na.rm = TRUE)
data$Unit <- ifelse(data$Type == 'u', 1, 0)
data$Townhouse <- ifelse(data$Type == 't', 1, 0)
data$Southern_Metro <- ifelse(data$Regionname == 'Southern Metropolitan', 1, 0)
data$X1 <- data$Rooms_c
data$X2 <- data$Distance_c
data$X3 <- data$Unit
data$X4 <- data$Townhouse
data$X5 <- data$Southern_Metro
data$X6 <- data$Bedroom2
data$X7 <- data$Bathroom
data$X8 <- data$Car
data$X9 <- data$Landsize
data$X10 <- data$BuildingArea

#additive terms
data$X11 <- data$X1^2
data$X22 <- data$X2^2
data$X111 <- data$X1^3
data$X222 <- data$X2^3
data$X1111 <- data$X1^4
data$X2222 <- data$X2^4
```

```r
#interactive terms
data$X1X2 <- data$X1 * data$X2
data$X1X3 <- data$X1 * data$X3
data$X1X4 <- data$X1 * data$X4
data$X1X5 <- data$X1 * data$X5
data$X1X6 <- data$X1 * data$X6
data$X1X7 <- data$X1 * data$X7
data$X1X8 <- data$X1 * data$X8
data$X1X9 <- data$X1 * data$X9
data$X1X10 <- data$X1 * data$X10

data$X2X3 <- data$X2 * data$X3
data$X2X4 <- data$X2 * data$X4
data$X2X5 <- data$X2 * data$X5
data$X2X6 <- data$X2 * data$X6
data$X2X7 <- data$X2 * data$X7
data$X2X8 <- data$X2 * data$X8
data$X2X9 <- data$X2 * data$X9
data$X2X10 <- data$X2 * data$X10

data$X3X4 <- data$X3 * data$X4
data$X3X5 <- data$X3 * data$X5
data$X3X6 <- data$X3 * data$X6
data$X3X7 <- data$X3 * data$X7
data$X3X8 <- data$X3 * data$X8
data$X3X9 <- data$X3 * data$X9
data$X3X10 <- data$X3 * data$X10

data$X4X5 <- data$X4 * data$X5
data$X4X6 <- data$X4 * data$X6
data$X4X7 <- data$X4 * data$X7
data$X4X8 <- data$X4 * data$X8
data$X4X9 <- data$X4 * data$X9
data$X4X10 <- data$X4 * data$X10
```

```r
data$X5X6 <- data$X5 * data$X6
data$X5X7 <- data$X5 * data$X7
data$X5X8 <- data$X5 * data$X8
data$X5X9 <- data$X5 * data$X9
data$X5X10 <- data$X5 * data$X10

data$X6X7 <- data$X6 * data$X7
data$X6X8 <- data$X6 * data$X8
data$X6X9 <- data$X6 * data$X9
data$X6X10 <- data$X6 * data$X10

data$X7X8 <- data$X7 * data$X8
data$X7X9 <- data$X7 * data$X9
data$X7X10 <- data$X7 * data$X10

data$X8X9 <- data$X8 * data$X9
data$X8X10 <- data$X8 * data$X10

data$X9X10 <- data$X9 * data$X10

data$X126 <- data$X1 * data$X2 * data$X6
data$X145 <- data$X1 * data$X4 * data$X5
data$X235 <- data$X2 * data$X3 * data$X5

n <- length(Y)
```

```r
model_final <- lm(Y ~
                  X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 +
                  X11 + X22 + X111 + X222 + X1111 + X2222 +
                  X1X2 + X1X3 + X1X4 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 +
                  X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 +
                  X3X4 + X3X5 + X3X6 + X3X7 + X3X8 + X3X9 + X3X10 +
                  X4X5 + X4X6 + X4X7 + X4X8 + X4X9 + X4X10 +
                  X5X6 + X5X7 + X5X8 + X5X9 + X5X10 +
                  X6X7 + X6X8 + X6X9 + X6X10 +
                  X7X8 + X7X9 + X7X10 +
                  X8X9 + X8X10 +
                  X9X10 +
                  X126 + X145 + X235,
                 data = data)
summary(model_final)

#Additive model
model2 <- lm(Y ~ X1 + X2 + X3 + X4 + X5 + X6 + X11 + X22 + X111 + X222 + X1111 + X2222, data = data)
summary(model2)

#Interaction Model
model1 <- lm(Y ~
             X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 + X1X2 + X1X3 + X1X4 + X1X5 + X1X6 + X1X7
             X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 +
             X3X4 + X3X5 + X3X6 + X3X7 + X3X8 + X3X9 + X3X10 +
             X4X5 + X4X6 + X4X7 + X4X8 + X4X9 + X4X10 +
             X5X6 + X5X7 + X5X8 + X5X9 + X5X10 +
             X6X7 + X6X8 + X6X9 + X6X10 +
             X7X8 + X7X9 + X7X10 +
             X8X9 + X8X10 +
             X9X10 +
             X126 + X145 + X235, data = data)
summary(model1)
```

```r
library(ggplot2)
library(gridExtra)
library(scales)
# Predicted values
data$Y_pred_model1 <- predict(model1, newdata = data)
data$Y_pred_model2 <- predict(model2, newdata = data)


# Plot 1: Interaction Model
p1 <- ggplot(data, aes(x = X1, y = Y, color = as.factor(X5))) +
  geom_point() +
  geom_line(aes(y = Y_pred_model1), size = 1) +
  labs(title = "Interaction Model", x = "Rooms (centered)", y = "Price (Millions)", color = "Southern Metro") +
  scale_y_continuous(labels = label_number(scale = 1e-6, suffix = "M", prefix = "$")) +  # <-- millions format
  theme_minimal()

# Plot 2: Additive Model
p2 <- ggplot(data, aes(x = X1, y = Y, color = as.factor(X5))) +
  geom_point() +
  geom_line(aes(y = Y_pred_model2), size = 1) +
  labs(title = "Additive Model", x = "Rooms (centered)", y = "Price (Millions)", color = "Southern Metro") +
  scale_y_continuous(labels = label_number(scale = 1e-6, suffix = "M", prefix = "$")) +  # <-- millions format
  theme_minimal()

# Arrange side by side
grid.arrange(p1, p2, ncol = 2)
```

```r
#model selection
library(leaps)
# To remove linear dependencies
alias(lm(Y ~
          X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 +
          X11 + X22 + X111 + X222 + X1111 + X2222 +
          X1X2 + X1X3 + X1X4 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 +
          X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 +
          X3X4 + X3X5 + X3X6 + X3X7 + X3X8 + X3X9 + X3X10 +
          X4X5 + X4X6 + X4X7 + X4X8 + X4X9 + X4X10 +
          X5X6 + X5X7 + X5X8 + X5X9 + X5X10 +
          X6X7 + X6X8 + X6X9 + X6X10 +
          X7X8 + X7X9 + X7X10 +
          X8X9 + X8X10 +
          X9X10 +
          X126 + X145 + X235,
        data = data))
#subset selection
m3 <- regsubsets(Y ~
                 X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8 + X9 + X10 +
                 X11 + X22 + X111 + X222 + X1111 + X2222 +
                 X1X2 + X1X3 + X1X4 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 +
                 X2X3 + X2X4 + X2X5 + X2X6 + X2X7 + X2X8 + X2X9 + X2X10 + X3X5 + X3X7 + X3X8 + X3X9 + X3X10 +
                 X4X5 + X4X7 + X4X8 + X4X9 + X4X10 +
                 X5X6 + X5X7 + X5X8 + X5X9 + X5X10 +
                 X6X7 + X6X8 + X6X9 + X6X10 +
                 X7X8 + X7X9 + X7X10 +
                 X8X9 + X8X10 +
                 X9X10 +
                 X126 + X145 + X235, really.big = TRUE,
                data = data)
Leaps3<- summary(m3)
with(Leaps3,round(cbind(which,rsq,adjr2,cp,bic),3))
```

```r
# backward elimination
backward.reg <- step(model_final,direction="backward",k=log(n))
# forward selection
reg.null <- lm(Y ~ 1,data=data)
forward.reg <- step(reg.null,direction="forward",scope=list(upper=model_final,lower=reg.null))
summary(forward.reg)
#Stepwise regression
stepwise_model <- step(model_final, direction = "both")


# model fits
backward_model <- lm(Y ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X22 + X2222 +
                       X1X2 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 + X1X10 + X2X3 +
                       X2X4 + X2X6 + X3X5 + X5X6 + X6X7 + X6X8 + X6X9 + X6X10 +
                       X7X10, data = data)
summary(backward_model)
forward_model <- lm(Y ~ X5X10 + X7 + X2X6 + X3X7 + X1111 + X7X10 + X22 + X1X7 + X2X3 +
                      X222 + X2X4 + X5X6 + X8 + X4X10 + X2X8 + X2X9 + X5X8, data = data)
summary(forward_model)
stepwise_model <- lm(Y ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 + X22 + X111 +
                       X1111 + X2222 + X1X2 + X1X5 + X1X6 + X1X7 + X1X8 + X1X9 +
                       X1X10 + X2X3 + X2X4 + X2X6 + X2X8 + X3X5 + X3X9 + X4X5 +
                       X4X7 + X4X10 + X5X6 + X5X9 + X5X10 + X6X7 + X6X8 + X6X9 +
                       X6X10 + X7X10 + X8X9 + X126, data = data)
summary(stepwise_model)
anova(backward_model, forward_model, stepwise_model)

# taking stepwise model for the regression analysis
```

```r
# taking stepwise model for the regression analysis
# Residual analysis
# Normality assumption check
residuals <- resid(stepwise_model)
qqnorm(residuals)
qqline(residuals, col = "red")

# Shapiro-Wilk test for normality
shapiro.test(residuals)

#Linearity assumption check
plot(stepwise_model, which = 1)

#Homoscedasticity assumption check
plot(stepwise_model, which = 3)
library(lmtest)
bptest(stepwise_model)

plot(cooks.distance(stepwise_model),
     main = "Cook's Distance", ylab = "Cook's Distance")
abline(h = 4/(nrow(data) - length(coef(stepwise_model))), col = "red")

library(MASS)
boxcox(stepwise_model, lambda = seq(-2, 2, 0.1))

stepwise_model_log <- lm(log(Y) ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 +
                           X22 + X111 + X1111 + X2222 + X1X2 + X1X5 + X1X6 + X1X7 +
                           X1X8 + X1X9 + X1X10 + X2X3 + X2X4 + X2X6 + X2X8 +
                           X3X5 + X3X9 + X4X5 + X4X7 + X4X10 + X5X6 + X5X9 +
                           X5X10 + X6X7 + X6X8 + X6X9 + X6X10 + X7X10 + X8X9 + X126,
                         data = data)
summary(stepwise_model_log)
plot(stepwise_model_log, which = 3)
```

```r
stepwise_model_sqrt <- lm(sqrt(Y) ~ X1 + X2 + X5 + X6 + X7 + X8 + X9 + X10 + X11 +
                            X22 + X111 + X1111 + X2222 + X1X2 + X1X5 + X1X6 + X1X7 +
                            X1X8 + X1X9 + X1X10 + X2X3 + X2X4 + X2X6 + X2X8 +
                            X3X5 + X3X9 + X4X5 + X4X7 + X4X10 + X5X6 + X5X9 +
                            X5X10 + X6X7 + X6X8 + X6X9 + X6X10 + X7X10 + X8X9 + X126,
                          data = data)
plot(stepwise_model_log, which = 2)
summary(stepwise_model_sqrt)
AIC(stepwise_model, stepwise_model_log, stepwise_model_sqrt)


summary(stepwise_model_log)$coefficients[c("X1", "X2", "X11", "X22", "X111", "X1111", "X2222", "X1X2"), ]
summary(stepwise_model_log)$coefficients[c("X1", "X2"), ]
summary(stepwise_model_log)$coefficients[c("X2X3", "X2X4", "X3X5", "X3X9"), ]
summary(stepwise_model_log)$coefficients[c("X7", "X6X7", "X7X10"), ]
summary(stepwise_model_log)$coefficients[c("X1X5"), ]

pred_log <- predict(stepwise_model_log, newdata = data, interval = "prediction")

# Convert from log(price) to actual price
pred_price <- exp(pred_log)

# Add predictions back to the data
data_with_predictions <- cbind(data, pred_price)

# View first few results
head(data_with_predictions)
data$pred <- data_with_predictions
```

P.S – Most of the outputs are integrated in the document.